

taken to run the script is also recorded in the output.

```
$ cat batch.Rout
R version 4.1.0 (2021-05-18) -- "Camp Pontanezen"
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Platform: x86_64-pc-linux-gnu (64-bit)
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Type 'license()' or 'licence()' for distribution details.
  Natural language support but running in an English locale
```

```
R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.
Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
```

```
Type 'q()' to quit R.
```

```
[Previously saved workspace restored]
> bank <- read.csv(file="bank.csv", sep=";")
>
> head(bank)
  age      job marital education default balance housing
loan contact day
1 30 unemployed married primary no 1787 no no cellular 19
2 33 services married secondary no 4789 yes yes cellular 11
3 35 management single tertiary no 1350 yes no cellular 16
4 30 management married tertiary no 1476 yes yes unknown 3
5 59 blue-collar married secondary no 0 yes no unknown 5
6 35 management single tertiary no 747 no no cellular 23
```

```
  month duration campaign pdays previous poutcome y
1 oct       79         1    -1         0 unknown no
2 may      220         1   339         4 failure no
3 apr      185         1   330         1 failure no
4 jun      199         4    -1         0 unknown no
5 may      226         1    -1         0 unknown no
6 feb      141         2   176         3 failure no
>
> print(summary(bank))
      age      job      marital
education
Min. :19.00 Length:4521 Length:4521 Length:4521
1st Qu.:33.00 Class :character Class :character Class
:character
Median :39.00 Mode :character Mode :character Mode
:character
Mean :41.17
3rd Qu.:49.00
Max. :87.00
      default      balance      housing
```

```
loan
Length:4521 Min. : -3313 Length:4521
Length:4521
Class :character 1st Qu.: 69 Class :character Class
:character
Mode :character Median : 444 Mode :character Mode
:character
Mean : 1423
3rd Qu.: 1480
Max. :71188
      contact      day      month      duration
Length:4521 Min. : 1.00 Length:4521 Min. : 4
Class :character 1st Qu.: 9.00 Class :character 1st Qu.: 104
Mode :character Median :16.00 Mode :character Median : 185
Mean :15.92 Mean : 264
3rd Qu.:21.00 3rd Qu.: 329
Max. :31.00 Max. :3025
      campaign      pdays      previous      poutcome
Min. : 1.000 Min. : -1.00 Min. : 0.0000
Length:4521
1st Qu.: 1.000 1st Qu.: -1.00 1st Qu.: 0.0000 Class
:character
Median : 2.000 Median : -1.00 Median : 0.0000 Mode
:character
Mean : 2.794 Mean : 39.77 Mean : 0.5426
3rd Qu.: 3.000 3rd Qu.: -1.00 3rd Qu.: 0.0000
Max. :50.000 Max. :871.00 Max. :25.0000
      y
Length:4521
Class :character
Mode :character
```

```
>
> proc.time()
      user system elapsed
0.160 0.036 0.144
```

You can also use the *Rscript* command to execute the above script, as shown below:

```
$ Rscript batch.R
```

The R commands can also be written to a source file with the shebang (“#!”) operator. For example:

```
#!/usr/bin/env Rscript
print("From executable Rscript!");
```

The script can be executed as follows:

```
$ ./batch_print.R
[1] "From executable Rscript!"
```